

Demonstration of a Dynamic SD-KMS with Batch-Based Multipath Routing in a Field-Deployed QKD Network

K. Tsimvrakidis⁽¹⁾, A. Selentis⁽¹⁾, A. Papageorgopoulos⁽¹⁾, I. Papastamatiou⁽²⁾, K. Christodouloupoulos⁽¹⁾, G. T. Kanellos⁽¹⁾

⁽¹⁾ National and Kapodistrian University of Athens, Greece kontsim@di.uoa.gr

⁽²⁾ GRNET S.A. – National Infrastructures for Research and Technology, Athens, Greece

Abstract We present a dynamic Software-Defined Key Management System (SD-KMS), integrated with a custom batch-based Linear Programming (LP) routing engine to optimize dynamic key provisioning in Quantum Key Distribution (QKD) networks. The SD-KMS is experimentally validated across diverse operational scenarios in a field-deployed QKD network. ©2026 The Author(s)

Introduction

Over the past decade, Quantum Key Distribution (QKD) Networks have transitioned from isolated, point-to-point links into large-scale, complex topologies [1]. Despite this, the majority of demonstrated QKD networks remain operationally static, employing simple Key Management System (KMS) processes [2] [3] for handling the extraction, storage, routing and provisioning of the quantum keys. Software defined networking (SDN) concepts offer the required agility for integration with carrier grade and fully operational network deployments. Recent efforts have introduced Software-Defined KMS (SD-KMS) frameworks to enhance network efficiency and flexibility [4][10]. However, they typically rely on a centralized key routing algorithm that does not follow the real time conditions of the QKD network such as varying application demands, fluctuating key generation rates, and varying key buffer levels.

A QKDN routing algorithm defines the key relay paths and contributions to those for Secure Application Entities (SAE) pairs. Such algorithm is used either for static planning or dynamic provisioning. Dynamic algorithms can be classified along two dimensions. First, by provisioning timing: they are either reactive [11]-[13] provisioning keys on demand (maximizing key efficiency at the expense of latency), or *proactive*, buffering keys at the KMS (with inverse performance) [14]. Second, by request processing: they serve demands either sequentially [11],[13],[14] or in batches [12]. Beyond these operational choices, path diversity is equally crucial. While most rely on a single path for a SAE pair, recent studies have proven the value of multipath approaches for both security and key throughput [15] [16].

In this work, we present and experimentally evaluate a custom SD-KMS that features real-time network-wide monitoring and response capabilities integrated with a ‘just-enough’ proactive routing engine to offer dynamic network and enhanced load balancing through batched multipath routing optimization. Aligned with modern multi-layer QKDN architectures [17] [18], the SD-KMS integrates standardized interfaces to interact with the QKD physical layer, while it operates

as an open, extensive framework that actively supports external routing solvers, passive route injection, and an embedded Dijkstra routing engine as a default option. In addition, we propose, implement and integrate a novel batch key routing engine based on Linear Programming (LP) that periodically computes optimal solutions in polynomial time and adapts to fluctuating demands and generation rates while employing multipath routing for improved security and performance. The complete dynamic SD-KMS and LP-based key routing engine is experimentally validated on a field-deployed, 5-node, multi-vendor QKD ring network and reveals real-time responsiveness to a network evolution scenario which includes link failure, key generation fluctuations, and varying demand conditions.

Key Management System

The developed SD-KMS separates logical key orchestration from the physical quantum network hardware, as per ETSI GS QKD 015 [2]. A centralized Controller manages network-wide routing while distributed KMS agents (in dedicated servers per QKD node) perform cryptographic data-plane operations locally. The KMS agents natively implement the ETSI GS QKD 014 interface [19] as a southbound abstraction layer to fetch raw symmetric 256-bit keys from any underlying (even heterogeneous) QKD device into Quantum Key Pools (QKPs) located at the KMS agent in that node. To handle high concurrency during simultaneous key harvesting and serving, the KMS agents utilize non-blocking database caching to eliminate thread starvation and minimize operation latencies. For non-adjacent Secure Application Entities (SAEs), the SD-KMS carries out an Information-Theoretic Secure (ITS) multi-hop relay. Rather than applying classical encryption (e.g. AES) for key relaying, the KMS agents use the local QKD keys for hop-by-hop One-Time Pad. The source encrypts the application key via XOR with a key of the first QKD link, and the resulting ciphertext is forwarded to first relay node, which decrypts and re-encrypts it with a QKD key for the next link, etc. Thus, plain keys never traverse links, extending point-to-

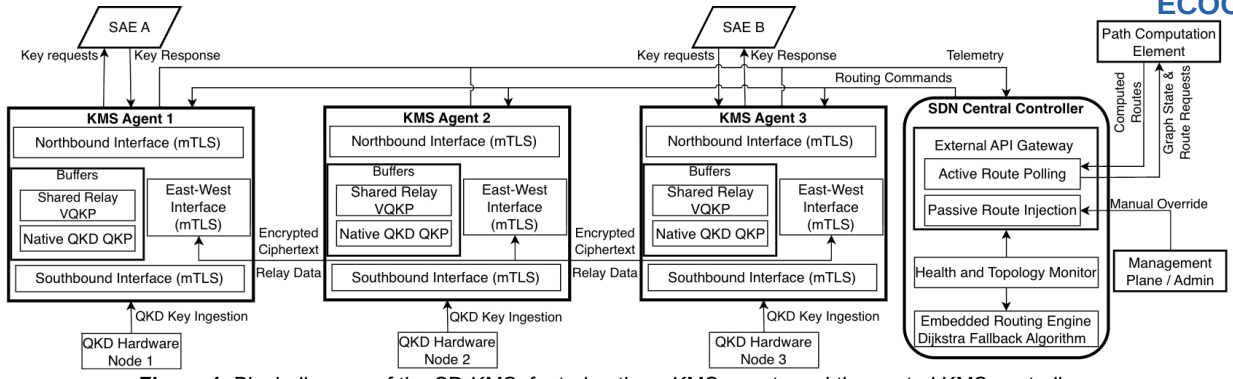


Figure 1. Block diagram of the SD-KMS, featuring three KMS agents and the central KMS controller.

point ITS guarantees to multi-node relays. Besides reactive serving, the SD-KMS can also employ proactive, SLA-driven key pooling (Virtual Quantum Key Pools – VQKP) between its non-neighboring SAEs. The Controller provides (QKP and VQKP levels) monitoring information, and the routing engine can schedule key relays in advance to pre-fill the VQKPs, so that SAEs experience zero key generation latency.

Additionally, to provide maximum operational flexibility and adaptability, the Controller is designed as a modular orchestration platform rather than a monolithic block. It manages key routing through three distinct operational mechanisms: i) An embedded, topology-aware Dijkstra engine that serves as a fallback algorithm to guarantee baseline service support. This engine calculates viable routes in the absence of an external routing engine. ii) An active polling interface: the controller acts as an active client that proactively pushes real-time network telemetry (e.g. QKP, VQKP levels, SKR and demand profiles) via RESTful APIs to a designated external optimization routing engine. This allows network operators to employ sophisticated algorithms that e.g. provide explicit, multipath routing instructions to aggregate QKD keys. iii) A passive route injection API: the controller acts as a passive server and exposes an administrative interface allowing third-party global orchestrators (routing engines, AI models, etc.) to override the internal or polling cycle and enforce custom, explicit routing commands directly to the network. This makes the SD-KMS framework open and future-proof.

System model and LP routing

In real world, key generation rates and demands fluctuate. A proactive pooling strategy does not anticipate such fluctuations, making mistakes that lead to inefficient key relays and blocking. On the other hand, a reactive, greedy strategy, called when a key request arrives, effectively adapts to dynamic network conditions, but results in increased delays. Furthermore, calculating

paths for only the current request causes path overlaps and blocking in dynamic scenarios.

To combine the best of both worlds, we propose a 'just-enough' proactive routing scheme. It stores sufficient keys in VQKPs to immediately serve dynamic demands, while maintaining QKPs to absorb SKR generation fluctuations. By leveraging the periodic operation of the SD-KMS the proposed algorithm operates in periods and processes demand batches to achieve global optimization, while also supporting multiple paths to increase performance and security.

We let N denote the set of nodes, L the set of links and A the set of Secure Application Entity (SAE) pairs. We consider the dynamic evolution over time in periods of duration T . For period t , we denote by G_{ij}^t the keys generated between adjacent QKD nodes (i, j) , $l_{ij} \in L$, by D_{sd}^t the sum of requested keys and by C_{sd}^t the keys consumed by SAE $a_{sd} \in A$ between node s and d within that period. These parameters are unknown at the start of period t . We consider period t successful if $C_{sd}^t = D_{sd}^t, \forall a_{sd} \in A$. Each node (KMS agent) hosts one QKP per adjacent QKD node and one VQKP for each node with an active SAE pair. For SAE pair a_{sd} between node s and d we denote by $p_{sd,k} \in P_{sd}$ the k -th relay path. We denote by $g_{sd,k}^t$ the keys consumed during period t from all QKP_{ij} of all links l_{ij} that are part of the k -th path for a_{sd} ($l_{ij} \in p_{sd,k}$) and are stored in $VQKP_{sd}$. Let $g_{sd}^t = \sum_k g_{sd,k}^t$ be the total sum of keys transferred to $VQKP_{sd}$ at period t over P_{sd} multiple paths. The (V)QKPs at the start of period t are:

$$\forall l_{ij} \in L: QKP_{ij}^{t+1} = QKP_{ij}^t - \sum_{a_{sd} \in A} \sum_{p_{sd,k} \in P_{sd} | l_{ij} \in p_{sd,k}} g_{sd,k}^t + G_{ij}^t \quad (1)$$

$$\forall a_{sd} \in A: VQKP_{sd}^{t+1} = VQKP_{sd}^t - C_{sd}^t + g_{sd}^t \quad (2)$$

Note that buffer levels QKP_{ij}^t and $VQKP_{sd}^t$ and for $t+1$ correspond to the start of the respective period, while all others ($g_{sd,k}^t, g_{sd}^t, G_{ij}^t, C_{sd}^t$) correspond to aggregated values over the period. The key-relay routing problem is defined as follows:

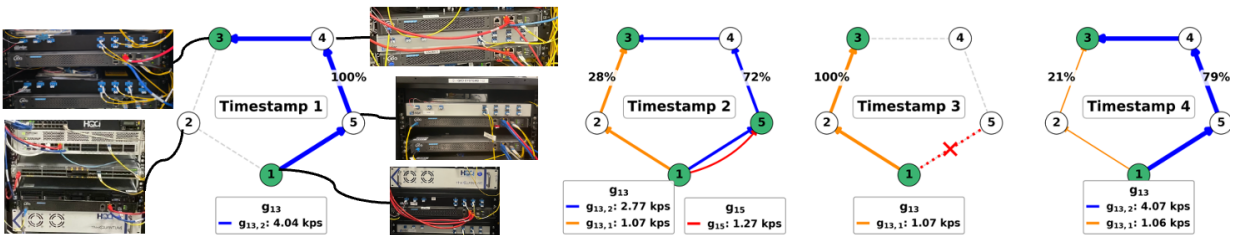


Figure 2. Dynamic adaption of network flows across four timestamps in the demonstrated scenario.

given buffer levels ($QKP_{ij}^t, VQKP_{sd}^t$) and demands D_{sd}^{t-1} provided by the Controller for the previous period, the algorithm decides $g_{sd,k}^t$ for the next period. Request processing can occur individually or in batches, using single or multiple paths.

To calculate the relaying paths $p_{sd,k}$ and key flows $g_{sd,k}^t, \forall a_{sd} \in A$ we developed a Multicommodity flow LP-based algorithm which considers demands in batches and provides multiple paths for each in polynomial time. Our objective is keeping each VQKP above a safety threshold ($VQKP_{low}^{sd}$) letting QKPs absorb generation fluctuations. At period t , if a VQKP remains above threshold, the requested keys match its $t-1$ consumption, whereas otherwise we proportionally increase key transfers to reach the threshold again. Due to space limitations, we omit the detailed description of the LP model.

Proof-of-Concept - Experimental Validation

To validate the performance and dynamic routing capabilities of the SD-KMS and LP-based key routing engine, we established a field-deployed, five-node QKD ring network across the metropolitan area of Athens, as illustrated in Fig. 2. The testbed incorporated a mix of QKD hardware, employing a polarisation-based DV-QKD pair from ThinkQuantum and time-bin/phase DV-QKD systems from ID Quantique. Although both vendors embed their own proprietary KMSs, our in-house SD-KMS was deployed as a unified overlay, interfacing with the underlying hardware and harvesting the QKD-generated keys through standardized ETSI GS QKD 014 API. At the application layer, although the network nodes included IPsec encryptors, the default key rotation was too infrequent to stress the QKD generation. Therefore, custom scripts were used to rapidly consume QKD keys via ETSI 014 requests.

The overall performance was validated in a dynamic network environment subject to key generation fluctuations, and varying demands and a link failure. Buffer capacities were constrained to

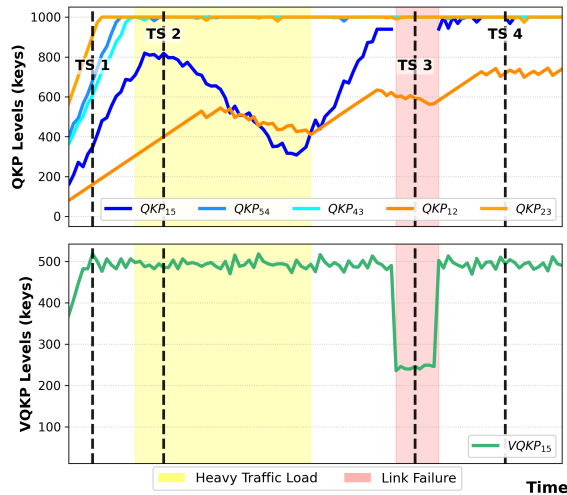


Figure 3. QKP(a) and VQKP(b) levels across the demonstrated scenarios.

1000 keys for QKPs, while the period T of the routing scheme was set to 15 sec.

In Fig. 3, we present the QKP and VQKP levels over time and in Fig. 2 we show the routes and flow rates (buffer level slopes) for four specific timestamps (TS). Note that we report key rates which can be translated to keys per period used in Section III for the given period of $T=15$ sec. We also suppress period counting notation (t) in the following. The network was initialized with all QKD links actively generating keys. The SD-KMS operated as a consumer to the embedded KMS of the QKD systems, therefore we define the resulting input key rate as our generation rate (G). While G_{23} , G_{34} , G_{45} , and G_{51} averaged between 4.00 and 4.5 keys per second (kps), G_{12} was relatively lower with an average of 1.07 kps as shown in Fig. 3.a (buffer level slopes). Shortly after key generation started, we activated a SAE pair between nodes 1 and 3 (a_{13}), which remained active throughout the demonstration. In the ring a_{13} had two paths: $p_{13,1}$ clockwise (1-2-3) and $p_{13,2}$ counterclockwise (1-5-4-3). We set the pool target $VQKP_{low}^{13} = 500$ keys and consumed with a rate of $C_{13} = 3$ kps. Our LP engine routed the requested keys through the counterclockwise path $p_{13,2}$ (1-5-4-3) due to its higher QKP capacity (TS 1). Subsequently, we activated an extra SAE pair (indicated by the yellow area in Fig. 3) between nodes 1 and 5 (a_{15}) with a consumption rate of $C_{15} = 7.5$ kps. As a result, the available keys for the pair a_{13} along $p_{13,2}$ decreased (suppressing $g_{13,2}$), and thus, the LP engine rerouted a portion of the relay flows to the clockwise path $g_{13,1}$, to maintain high key provisioning and prevent pool starvation (TS 2). Specifically, the flows reached $g_{13,1}=1.07$ kps and $g_{13,2}=2.77$ kps achieving a total key rate of $g_{13}=3.84$ kps, outperforming single-path solutions. After satisfying the a_{15} demand, we intentionally failed the link l_{15} , rendering the counterclockwise path $p_{13,2}$ unusable. The engine successfully rerouted the entire traffic to clockwise path $p_{13,1}$, maintaining continuity of service (TS 3). Lastly, upon link restoration, the engine routed keys through both paths, with $g_{13,2}=4.07$ kps and $g_{13,1}=1.06$ kps along paths 2 and 1, increasing $VQKP_{13}$ filling rate to a total of $g_{13}=5.13$ kps, to optimize served requests (TS 4). It adjusted the flows following generation fluctuations until the end of the demonstration.

Conclusions

We developed a dynamic SD-KMS and interfaced it with a custom batch-based multi-path routing engine based on LP, tailored to optimize the dynamic key provisioning in QKDNs. The SD-KMS and the LP-based routing engine was successfully validated through a field-deployed, 5-node ring QKD network.

Acknowledgements

This work was partially funded by EU projects SEEWQCI (GA101249531) and ORQUESTRA (GA101224573).

References

- [1] Y. Cao, Y. Zhao, Q. Wang, J. Zhang, S. X. Ng, and L. Hanzo, "Evolution of Quantum Key Distribution Networks: On the Road to the Q-Internet," in *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 839-894, 2022.
- [2] ETSI, "Quantum Key Distribution (QKD); Control Interface for Software Defined Networks," ETSI GS QKD 015 V2.1.1, 2022.
- [3] ITU-T, "Quantum key distribution networks - Control and management," Recommendation ITU-T Y.3804, 2020.
- [4] A. Aguado, V. Lopez, D. R. Lopez, M. Peev, A. Poppe, and V. Martin, "The engineering of software-defined quantum key distribution networks," *IEEE Communications Magazine*, vol. 57, no. 7, pp. 20-26, 2019.
- [5] R. S. Tessinari, A. Bravalheri, E. Hugues-Salas, R. Collins, D. Aktas, R. S. Guimaraes, O. Alia, J. Rarity, G. T. Kanellos, R. Nejabati, and D. Simeonidou, "Field trial of dynamic dv-qkd networking in the sdn-controlled fully-meshed optical metro network of the bristol city 5g uk test network," in *45th European Conference on Optical Communication (ECOC 2019)*, IET, 2019, pp. 1-4.
- [6] O. Alia, R. S. Tessinari, E. Hugues-Salas, G. T. Kanellos, R. Nejabati, and D. Simeonidou, "Dynamic DV-QKD Networking in Trusted-Node-Free Software-Defined Optical Networks," *Journal of Lightwave Technology*, vol. 40, no. 17, pp. 5816-5824, 2022.
- [7] S. Gupta, I. Agarwal, V. Mogiligidda, R. K. Krishnan, S. Chennuri, D. Aggarwal, A. Hoodati, S. Cooper, Ranjan, M. B. Sheik, K. M. Bhavya, M. Hegde, M. N. Krishna, A. K. Chauhan, M. Korrapati, S. Singh, J. B. Singh, S. Sud, S. Gupta, S. Pant, Sankar, N. Agrawal, A. Ranjan, P. Mohapatra, T. Roopak, A. Ahmad, M. Nanjunda, and D. Singh, "ChaQra: a cellular unit of the Indian quantum network," *Scientific Reports*, vol. 14, no. 1, p. 16752, 2024.
- [8] V. Martin, J. P. Brito, L. Ortiz, R. B. Mendez, J. S. Buruaga, R. J. Vicente, A. Sebastián-Lombraña, D. Rincon, F. Perez, C. Sanchez, M. Peev, H. H. Brunner, F. Fung, A. Poppe, F. Fröwis, A. J. Shields, R. I. Woodward, H. Griesser, S. Roehrich, F. De La Iglesia, C. Abellan, M. Hentschel, J. M. Rivas-Moscoco, A. Pastor, J. Folgueira, and D. R. Lopez, "MadQCI: a heterogeneous and scalable SDN-QKD network deployed in production facilities," *npj Quantum Information*, vol. 10, no. 1, p. 80, 2024.
- [9] R. B. Mendez, J. Ballesta, R. Querol, C. Escribano, P. Salas, A. Pastor, D. Lopez, V. Martin, and J. P. Brito, "Coexistence of SDN Management Protocols in QKD Networks," *2025 IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC)*, pp. 169-173, 2025.
- [10] A. Sanz, A. Atutxa, D. Franco, J. Astorga, E. Jacob, and D. López, "Toward Quantum-Safe Scalable Networks: An Open, Standards-Aware Key Management Framework," *IEEE Network*, vol. PP, no. 99, pp. 1-10, 2025.
- [11] Q. Zhang, O. Ayoub, A. Gatto, J. Wu, F. Musumeci, and M. Tornatore, "Routing, Channel, Key-Rate, and Time-Slot Assignment for QKD in Optical Networks," in *IEEE Transactions on Network and Service Management*, vol. 21, no. 1, pp. 148-160, 2024.
- [12] Y. Cao, Y. Zhao, C. Colman-Meixner, X. Yu, and J. Zhang, "Key on demand (KoD) for software-defined optical networks secured by quantum key distribution (QKD)," *Optics Express*, vol. 25, no. 22, pp. 26453-26467, 2017.
- [13] S. Xu, Y. Zhao, L. Huang, and C. Qiao, "High-Efficient Quantum Key Distribution With Routing and Photon Source Provisioning," in *IEEE/ACM Transactions on Networking*, vol. 34, pp. 2119-2134, 2026.
- [14] C. Lee, Y. Kim, K. Shim, and W. Lee, "Key-count differential-based proactive key relay algorithm for scalable quantum-secured networking," in *Journal of Optical Communications and Networking*, vol. 15, no. 5, pp. 282-293, 2023.
- [15] M. Ahmadian, F. Arpanaei, J. C. Hernandez-Hernandez, R. Lin, and P. Monti, "Enhancing the reliability of multipath QKD over multi-band systems," in *Journal of Optical Communications and Networking*, vol. 17, no. 12, pp. 1105-1116, December 2025.
- [16] M. Wang, J. Li, K. Xue, R. Li, N. Yu, Y. Li, Y. Liu, Q. Sun, and J. Lu, "A segment-based multipath distribution method in partially-trusted relay quantum networks," *IEEE Communications Magazine*, vol. 61, no. 12, pp. 184-190, 2023.
- [17] N. Makris, A. Papageorgopoulos, K. Tsimvrakidis, P. Konteli, Y. Gautier, M. Terenziani, E. Daudin, D. Ntoulas, T. Fragkioudakis, I. Meletios, M. Mosca, D. Hobbs, T. Rosati, I. Papastamatiou, O. Prnjat, K. Koumantaros, D. Mitropoulos, J.-R. Morax, B. Huttner, O. K. Christodouloupoulos, G. T. Kanellos, and D. Syvridis, "Field Demonstration of a Fully Managed, L1 Encrypted 3-Node Network with Hybrid Relayed-QKD and Centralized Symmetric Classical Key Management," *2024 50th European Conference on Optical Communication (ECOC)*, Frankfurt, Germany, 2024, pp. 1-4.
- [18] K. Tsimvrakidis, A. Papageorgopoulos, S. Laschet, F. Kutschera, P. James, L. Torresetti, S. Ramacher, N. Makris, P. Konteli, J. S. Buruaga, R. J. Vicente, Y. Lobaina, J. P. Brito, P. Papapetropoulos, I. Papastamatiou, H. Hübel, V. Martin, and G. T. Kanellos, "Interoperable Multi-Vendor, Multi-Domain, Multi-Layer, Multi-Service Quantum Key Distribution Network Field-Trial," in *International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2026.
- [19] ETSI, "Quantum Key Distribution (QKD); Protocol and data format of REST-based key delivery API," ETSI GS QKD 014 V1.1.1, 2019.